

# CHEMISTRY-CONNECT

## S5 Chemistry Scenario based Items

**Item 1:** An oil pipeline engineer made a mistake while attempting to locate a leak on an oil pipeline. He accidentally added 78g of radioactive substance at a pumping station. This set off alarms indicating the levels of radioactivity in the oil were very dangerous. The pipe had to be shut down. The facility and the nearby community was vacated due to the dangerously high levels of radiation.

The environmental authorities then set up meters that measured the levels of radioactivity in the area. They got the data below:

|                              |    |     |     |     |     |     |
|------------------------------|----|-----|-----|-----|-----|-----|
| Time (days)                  | 0  | 10  | 20  | 30  | 40  | 50  |
| Activity (counts per second) | 21 | 8.9 | 3.8 | 1.6 | 0.7 | 0.3 |

The government is claiming that the area will be safe again if the amount of the radioactive substance in the oil reduces to 5% and that this will happen in two weeks.

### Task

As a student of Chemistry and showing clear working, using the information provided, respond to the claims of the government.

(Use the graph paper attached)

**ITEM 2:** At **Mbarara Plastics Ltd**, workers often mix different types of plastic waste during recycling. This causes some recycled containers to melt irregularly or produce offensive odours. The quality control officer suspects that the plastics belong to **different classes of organic compounds** containing varied **functional groups** such as esters, alcohols, and alkanes. The company invites A-Level Chemistry learners to analyse and help classify the compounds for proper recycling.

**Task:** As a Chemistry learner,

- (a) Identify the **main functional groups** likely present in the different plastics and relate them to their chemical properties.
- (b) Explain how the **type of bonding in carbon chains** influences melting behaviour during recycling.
- (c) Using molecular examples, show how **isomerism** could lead to plastics with different melting points even if they have the same molecular formula.
- (d) Suggest **two ways** the company could improve sorting and recycling based on organic compound classification.

**Item 3:** A student intern at **Kampala Pharmaceutical Research Centre** mistakenly labels some compound samples with only partial names such as “propyl alcohol” and “methyl propanoate.” The laboratory supervisor asks the student to recheck the **IUPAC names** and verify the structures before the compounds are used in synthesis. The supervisor requests A-Level Chemistry learners to assist the intern in identifying and naming the compounds correctly.

**Task:** As a Chemistry learner,

- (a) Explain the **importance of IUPAC nomenclature** in pharmaceutical formulation.

- (b) Write the **full structural formula** of the compound named “propyl alcohol” and identify its functional group.
- (c) Deduce the **IUPAC name** of a compound with the molecular formula  $C_3H_6O_2$  that contains an ester group.
- (d) Suggest **two ways** proper IUPAC naming prevents errors in drug manufacturing and storage.

**Item 4:** Barium fluoride and magnesium fluoride have various applications in optics due to their transparency over a wide range of wavelengths. A company is to synthesize one of these compounds to be used as a raw material for other optics related devices. The design of their manufacturing process requires that they select to produce the compound that gives out the less heat as it forms from its gaseous component ions.

As a student of Chemistry consider the data below

$MgF_2$  : Standard heat of formation =  $-1124 \text{ kJ/mol}$ ; standard heat of sublimation of magnesium =  $+148 \text{ kJ/mol}$ ,  $IE_1(Mg) = +738 \text{ kJ/mol}$ ,  $IE_2(Mg) = +1451 \text{ kJ/mol}$ , Bond dissociation energy ( $F_2$ ) =  $+159 \text{ kJ/mol}$ , First electron affinity (F) =  $-328 \text{ kJ/mol}$ .

$BaF_2$  : Standard heat of formation =  $-1207 \text{ kJ/mol}$ , Standard heat of sublimation (Ba) =  $+180 \text{ kJ/mol}$ ;  $IE_1(Ba) = +503 \text{ kJ/mol}$ ;  $IE_2(Ba) = +965 \text{ kJ/mol}$ ,  $EA(F) = -328 \text{ kJ/mol}$ .

**Task:**

- Use your knowledge of energy cycles and determine which of the two compounds will be the more suitable.
- Help the company chemical engineer understand the stability difference between the two compounds to address durability concerns in optical devices they are to make.

**Item 5:** Study the data below about the first ten successive energies (in kJ per mole) required to remove the first ten electron from the gaseous atom and then ions of an element X and another one Y.

|   |       |        |        |       |       |       |       |       |       |        |
|---|-------|--------|--------|-------|-------|-------|-------|-------|-------|--------|
| X | 495.8 | 4562   | 6910.3 | 9543  | 13354 | 16613 | 20117 | 25496 | 28932 | 141362 |
| Y | 577.5 | 1816.7 | 2744.8 | 11577 | 14842 | 18379 | 23326 | 27465 | 31853 | 38473  |

**Tasks**

As a student of Chemistry

- Explain the category of the process involved in the removal of the first electron and state any two factors that can affect the amount of the energy required.
- Suggest the acceptable formulae of the chlorides of these elements and explain your reasons
- Which of the chlorides you have given in (b) above is more likely to dissolve in paraffin? Give a concise explanation of your reasons.

**Item 6:** A local beverage factory produces fruit drinks using artificial flavours. During quality testing, chemists discovered that two samples labelled “banana flavour” and “pear flavour” had the same **molecular formula** ( $C_4H_8O_2$ ) but produced completely different aromas. This led the factory to investigate **isomerism and functional groups** in organic compounds. The research manager invites Chemistry learners to explain why compounds with the same formula can behave differently.

**Task:**

As a Chemistry learner,

- (a) Explain why compounds with the same **molecular formula** can have **different smells or properties**.
- (b) Using **structural examples**, show how **functional group isomerism** can exist between an **acid** and an **ester** with formula  $C_4H_8O_2$ .
- (c) State one **test** that could distinguish between the two flavour compounds.
- (d) Suggest **two benefits** of understanding isomerism for the **food or perfume industry**.

**Item 7:** In many Ugandan homes, people cook using **charcoal, kerosene, or LPG (propane and butane gases)**. The **Ministry of Energy** is investigating why LPG burns more cleanly and efficiently than other fuels. Scientists suspect that the **type of hydrocarbon bonding and molecular structure** determine flame characteristics and pollution levels. A-Level Chemistry learners are invited to analyse the fuels and advise the ministry.

**Task:** As a Chemistry learner,

- (a) Identify the **hydrocarbon class** of LPG and relate its bonding to its **clean combustion** behaviour.
- (b) Explain why **kerosene** has a **higher boiling point** than LPG, relating it to intermolecular forces.
- (c) Account for the **formation of soot** in kerosene compared to LPG.
- (d) Propose **two scientific measures** the ministry can promote to improve energy efficiency and air quality.

**Item 8:** At **Kampala Polymer Works**, engineers manufacture plastic bottles from **ethene gas**. A recent system fault allowed **bromine vapour** and **hydrogen chloride** to enter the polymerisation chamber, producing brittle, coloured plastics instead of clear ones. The management consults A-Level Chemistry learners to determine the **chemical cause** and how to prevent recurrence.

**Task:** As a Chemistry learner,

- (a) Explain why **ethene** reacts readily with bromine and hydrogen chloride.
- (b) State the **observation** confirming the reaction between ethene and bromine.
- (c) Distinguish between **addition reactions** and **polymerisation** in ethene.
- (d) Suggest **two technical adjustments** that would restore correct polymer production.

**Item 9.** In urban areas like **Kampala**, incomplete combustion of petrol and diesel leads to the release of **unburnt hydrocarbons** and **carbon monoxide**. Air-quality inspectors discovered that most of the hydrocarbons are **alkanes** from fuel evaporation and **alkenes** from partial oxidation in engines. The **National Environment Authority (NEMA)** has asked Chemistry learners to analyse the chemical basis of these emissions and suggest solutions.

**Task:** As a Chemistry learner,

- (a) Explain why **alkanes** are generally less reactive in the atmosphere compared to alkenes.
- (b) Describe how **ultraviolet light** initiates the chlorination of alkanes in the presence of chlorine gas.
- (c) Use a **reaction mechanism** to illustrate the **free-radical substitution** of methane by chlorine.
- (d) Suggest **two ways** government policy could reduce hydrocarbon emissions from vehicles.

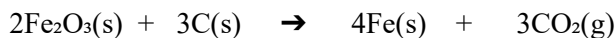
**Item 10:** Farmers in **Kayunga District** discovered that bananas ripen faster when stored near ripe fruits. Research by the **National Agricultural Research Organisation (NARO)** showed that ripe fruits emit **ethene gas**, which accelerates ripening of others. The farmers seek help to understand the chemistry behind this observation.

**Task:** As a Chemistry learner,

- (a) Identify the **hydrocarbon type** of ethene and explain why it acts as a **reactive gas**.
- (b) Show, using a **reaction mechanism**, how ethene reacts with **bromine water** to form a saturated compound.

- (c) Explain how this reactivity relates to its **effect on fruit ripening**.  
(d) Suggest **two safe methods** farmers could use to control or regulate fruit ripening.

**Item 11:** A factory uses charcoal that comes in 65kg bags each at 65000 UgX in smelting process to produce iron. The reaction occurs as below:



A procurement and accounts planning officer has come to you from the factory with a problem. They need your help. They need an estimate on how much it will cost in charcoal to produce 150 tonnes of iron and how much they will pay in environmental protection tax if the rate is 78000 per tonne of carbon dioxide released into the atmosphere.  
(Take atomic masses Fe = 56, C = 12 and O = 16)

**Task**

Explain the category of reactions the factory uses in producing iron.

- With clear working, provide answers to the officer from the factory. (Assume the charcoal is 100% carbon)
- With clear working, provide answers to the officer from the factory. (Assume the charcoal is 100% carbon)
- Suggest ways in which the factory can minimise its contribution to environmental pollution.

**Item 12:** It is common knowledge in Chemistry that elements in a group of the Periodic Table have the similar physical and chemical properties. A closer study of properties of elements in the first three groups reveals that the first element in each group is similar in physical and chemical properties to the element in the next group to the right but in the next period below:

Tasks Using your knowledge of periodic trends in the Periodic Table

- Identify the phenomenon described above.
- Describe the similarities in physical and chemical properties between any two elements that show the phenomenon you have identified in (a) above.
- Describe any two ways in which the implications of the phenomenon you have identified in (a) above are applied in industry.

**Item 13:** At **Buganda Metal Works**, technicians use **acetylene (C<sub>2</sub>H<sub>2</sub>)** mixed with oxygen for metal cutting and welding. However, the manager complains of **irregular flame temperatures** and soot formation. The company invites Chemistry learners to evaluate the chemistry of acetylene and its burning process.

**Task:** As a Chemistry learner,

- Identify the **type of bonding** present in acetylene and relate it to its **combustion behaviour**.
- Explain why incomplete burning of acetylene produces a **sooty flame**.
- Illustrate with a **reaction mechanism** how acetylene undergoes **addition with hydrogen** to form ethane.
- Suggest **two industrial precautions** to ensure safe and efficient use of acetylene.

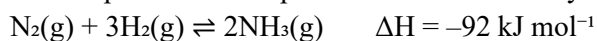
**Item 6:** At **Uganda Petroleum Refinery**, chemists are exploring how to produce **ethanol** from crude-oil fractions rather than from sugarcane fermentation. They propose converting **ethene** into ethanol using **acid-catalysed hydration**. Before approval, the refinery consults A-Level Chemistry learners to explain the underlying reactions.

**Task:**

As a Chemistry learner,

- (a) Outline the **type of reaction** that converts ethene to ethanol.
- (b) Use a **reaction mechanism** to show how **acid-catalysed addition of water** to ethene occurs.
- (c) Explain how **temperature and catalyst choice** influence yield and rate in this process.
- (d) Suggest **two advantages** of this petrochemical method over fermentation for large-scale ethanol production.

**Item 14:** At AgroGrow Uganda Ltd, ammonia is produced by reacting nitrogen and hydrogen gases under high pressure and temperature in the presence of an iron catalyst:

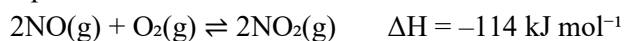


Recently, engineers observed a drop in ammonia yield, although gas supply was constant. Some suggest raising temperature to increase production speed, while others fear this might reduce yield. The management invites Advanced-Level Chemistry learners to analyse the system and recommend practical solutions for efficient operation at minimal cost.

**Task:** As a Chemistry learner,

- a) Identify the type of equilibrium in the process and one sign that shows it has been established.
- b) Describe how temperature influences ammonia yield, referring to the energy change in the equation.
- c) Using the data below, calculate the value of  $K_p$  and interpret what it indicates about the system:  
 $p(\text{N}_2) = 120 \text{ kPa}$ ,  $p(\text{H}_2) = 360 \text{ kPa}$ ,  $p(\text{NH}_3) = 80 \text{ kPa}$ .
- d) Suggest two adjustments the company could make to improve yield while keeping production costs low.

**Item 15:** In Kampala, thick brown fumes often appear during traffic jams. According to the National Environment Management Authority (NEMA), the color results from nitrogen dioxide gas produced in vehicle engines through the equilibrium:



When engines become hotter, more  $\text{NO}_2$  is formed, worsening air pollution. Engineers intend to install platinum-coated catalytic converters to control these gases and have requested Chemistry learners to evaluate how equilibrium conditions influence  $\text{NO}_2$  production and propose pollution-control measures.

**Task:** As a Chemistry learner,

- a) State what happens to the forward and reverse reactions when engine temperature rises.
- b) Describe how increasing oxygen concentration affects the equilibrium composition.
- c) Explain the role of the platinum catalyst and how it differs from changing temperature or pressure.
- d) Suggest two practical strategies (chemical or policy-based) to reduce nitrogen-oxide pollution in urban areas.

**Item 16:** At **Kampala Chemical Works**, a drum of **chloroethane** burst near a heater. Workers experienced dizziness, and the safety officer noticed a strong, irritating odour. She suspects that the solvent decomposed under heat to release harmful gases. The company calls in Chemistry students to analyse what happened and advise on safe handling.

**Task:**

As a Chemistry learner,

- (a) Using the situation above, explain how **heat exposure** could have broken the carbon–halogen bond in chloroethane.
- (b) Suggest what **toxic product** might form during such decomposition and how it relates to the observed odour.
- (c) Outline the **mechanism** by which chloroethane would react with aqueous alkali if it leaked into water.
- (d) Propose **two corrective safety measures** the factory should adopt to prevent a similar incident.

**Item 17:** During a practical at **Light Vocational Secondary School**, one group heats **bromoethane** with **aqueous KOH**, while another uses **alcoholic KOH**. The two groups obtain completely different products. Their teacher asks Chemistry learners to analyse the reason behind this outcome and suggest how to get consistent results next time.

**Task:** As a Chemistry learner,

- (a) Explain, using the information above, why the two groups obtained **different organic products**.
- (b) Using mechanisms, show how **bromoethane reacts** under each set of conditions.
- (c) Identify which group's method produced **ethene** and justify your answer.
- (d) Suggest **two experimental controls** the class could use to ensure only one product forms.

**Item 18:** At **Kabale Steel Works**, engineers use carbon monoxide to reduce iron(III) oxide to metallic iron. The process consumes large amounts of coal, and management wants to know if there's a **more energy-efficient route** to achieve the same result. They consult Chemistry learners to assess the feasibility of adjusting furnace conditions using thermochemical data.

**Task:** As a Chemistry learner,

- (a) Identify the **type of enthalpy change** involved in the reduction of iron(III) oxide.
- (b) Explain how **temperature** affects the reaction's **enthalpy profile and activation energy**.
- (c) Use your understanding of **energy diagrams** to suggest how catalysts could make the process more efficient.
- (d) Recommend **two measures** the plant could adopt to minimise energy waste during reduction.

**Item 19 .** A factory plans to produce small amounts of carbon dioxide by thermally decomposing the any one of the carbonates : magnesium carbonate, calcium carbonate or barium carbonate in an electric heater. They need to achieve this with the least expenditure on electricity and minimum side effects from handling.

**Tasks:** Consider the data below:

| Cation           | Size (pm) |
|------------------|-----------|
| Mg <sup>2+</sup> | 72        |
| Ca <sup>2+</sup> | 100       |
| Ba <sup>2+</sup> | 135       |

| Carbonate         | Decomposition temperature (°C) |
|-------------------|--------------------------------|
| MgCO <sub>3</sub> | 350                            |
| CaCO <sub>3</sub> | 840                            |
| BaCO <sub>3</sub> | 1350                           |

- a) Help the factory management (i) make a choice.
  - ii) Understand the chemistry principles of bonding and structure that make the choice you have made in (i) the most suitable.
- b) Describe the possible environmental impacts of this industrial process and suggest mitigation measures.

**\*END\***

*“Enjoy the beauty of Chemistry”*